# Replay-Aware CVRP Benchmark Report

## Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_no_replay.

## Run Metadata

- run_name: run_20260513_192724_i
- started_at_local: 2026-05-13 19:27:24
- finished_at_local: 2026-05-13 19:30:16
- duration_hhmm: 00:03
- duration_seconds: 171.787
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

## Condition Comparison

| Condition | Replay | Selection | Compression | Final CVRPLIB Gap | Final Synthetic Gap | Final Transfer Gap | Mean Accepted Novelty | Mean Complexity | Adaptation Efficiency |
|---|---|---|---|---|---|---|---|---|---|
| cvrp_no_replay | none | score_only | False | 0.234747 | 0.005376 | 0.132804 | 0.679487 | 0.78 | 0.036792 |
| cvrp_random_replay | random | score_only | False | 0.240934 | 0.005376 | 0.136242 | 0.911469 | 0.58 | 0.049481 |
| cvrp_failure_replay | failure | score_only | False | 0.238632 | 0.005376 | 0.134963 | 0.0 | 0.58 | 0.0 |
| cvrp_failure_replay_compression | failure | novelty_gate | True | 0.235906 | 0.005376 | 0.133448 | 0.899749 | 0.58 | 0.050162 |

## Condition Notes

### cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.234747, synthetic 0.005376, combined 0.132804.

- Accepted-epoch count 2, mean accepted code novelty 0.679487, and final complexity 0.78.
- Adaptation efficiency 0.036792 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.234747 across 5 instances; family means: A=0.151832, B=0.219154, E=0.28215, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

### cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.240934, synthetic 0.005376, combined 0.136242.
- Accepted-epoch count 2, mean accepted code novelty 0.911469, and final complexity 0.58.
- Adaptation efficiency 0.049481 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.240934 across 5 instances; family means: A=0.115183, B=0.229914, E=0.328215, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

### cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.238632, synthetic 0.005376, combined 0.134963.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.238632 across 5 instances; family means: A=0.143979, B=0.229914, E=0.287908, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

### cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.235906, synthetic 0.005376, combined 0.133448.

- Accepted-epoch count 2, mean accepted code novelty 0.899749, and final complexity 0.58.
- Adaptation efficiency 0.050162 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.235906 across 5 instances; family means: A=0.13438, B=0.225192, E=0.287908, P=0.306859.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 589, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.286026}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

## Judge Appendix

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# CVRP Replay-Aware Benchmark Suite Analysis

## Summary of Main Metrics (Lower gaps are better):

| Condition | CVRPLIB Gap | Synthetic Gap | Transfer Gap | Code Novelty | Complexity | Replay Mode |
|---|---|---|---|---|---|---|
| cvrp_no_replay | 0.2347 | 0.0054 | **0.1328** | 0.6795 | 0.78 | none |
| cvrp_random_replay | 0.2409 | 0.0054 | 0.1362 | **0.9115** | 0.58 | random |
| cvrp_failure_replay | 0.2386 | 0.0054 | 0.1350 | 0.0 | 0.58 | failure |
| cvrp_failure_replay_compression | 0.2359 | 0.0054 | 0.1334 | 0.8997 | 0.58 | failure + compression |

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## Optimality Gaps and Transfer

- The **no replay** condition (cvrp_no_replay) achieves the best held-out CVRPLIB gap (0.2347) and best synthetic holdout gap (0.0054).
- It also yields the lowest mean transfer gap (0.1328), outperforming all replay conditions.
- The replay conditions (random, failure, failure+compression) show slightly higher CVRPLIB gaps (~0.236 to 0.241) and transfer gaps (~0.133 to 0.136).
- Differences in synthetic gaps are negligible across conditions (all 0.0054).

## Code Novelty vs. Transfer

- Random replay and failure+compression replay increase **code novelty** substantially (~0.9 mean novelty), compared to no replay (0.68) and failure replay (0.0).

- Despite higher novelty, these replay modes do not improve transfer metrics and actually perform slightly worse than no replay.
- Failure replay condition with zero code novelty performs poorly on transfer compared to no replay.
- Thus, increased code novelty from replay modes does **not translate to better algorithmic transfer** or optimality gap improvements.

## Replay Mode Effects

- **No replay** (none) yields the best overall transfer and held-out gap metrics at the cost of higher complexity (0.78).
- **Random replay** trades off slight degradation in transfer/held-out gaps for significantly lower complexity (0.58) and higher novelty.
- **Failure replay** (no novelty) and **failure replay with compression** (high novelty) show no meaningful benefit in transfer metrics compared to no replay, despite replaying failure cases.
- Compression pressure in failure replay does not produce transfer gains.

## Complexity and Adaptation Efficiency

- No replay shows highest complexity (0.78), with replay modes consistently at 0.58 complexity.
- Adaptation efficiency is highest with failure+compression replay (0.050), followed by random replay (0.049), but this does not yield optimality gains.
- No replay has lowest adaptation efficiency (~0.037) but best transfer/held-out gap metrics.

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# Conclusions

- **Best condition for generalization and transfer is no replay (cvrp_no_replay)**, judged by held-out CVRPLIB gaps and synthetic gaps.
- Replay modes increase code novelty but do not improve—and may slightly degrade—transfer performance.
- Compression-aware replay does not deliver transfer advantage vs. simple no replay, despite efforts to preserve novelty.
- Algorithmic improvements should prioritize reducing optimality gaps on held-out CVRPLIB benchmarks over maximizing code novelty.
- Replay conditions alter behavioral profiles and reduce complexity but without transfer gains.
- Claims of algorithmic invention based on code novelty alone are not substantiated by transfer or gap improvements here.

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# Recommendation

For applications prioritizing transfer and solution quality, **prefer no replay training conditions**. Use replay modes only if complexity reduction or controlled novelty is specifically desired, acknowledging potential slight loss in optimality and transfer performance.